



FIFA WORLD CUP 2026
USA.CANADA.MEXICO

FIFA World Cup 2026 Sponsorship Analytics Platform
(AI-Powered Business Intelligence & Cloud Analytics Solution)



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Executive Summary

Technical Overview

The platform incorporates multi-source data integration, Python ETL data flows, Snowflake cloud data warehousing, SQL analysis, Power BI executive data visualization, and OpenAI conversational AI analysis into an integrated analytical environment.

Project Objectives

The first and foremost aim of this project is to analyze sponsorship performance in different FIFA World Cup regions and discover high-opportunity markets through revenue analysis, operating efficiency metrics, customer engagement metrics, and market opportunity scoring models.

In addition, the system is designed to facilitate communication between business users and analytical data through interactive dashboarding and natural-language AI queries. Power BI dashboards offer executives KPI reporting and market strategy visualization, while the Streamlit AI assistant allows users to dynamically generate SQL queries and access business insights from Snowflake.

Major Project Achievements

- *Centralized Snowflake cloud data warehouse implementation*
- *Multi-source data integration via Python ETL pipelines*
- *Executive Power BI dashboard with interactive analytics*
- *Leveraging OpenAI GPT models for generating SQL queries using AI technology*
- *Creation of a conversation-based analytics assistant utilizing Streamlit*
- *Application of strategic opportunity analysis in FIFA sponsorship*

The proposed solution shows how cloud-based analytics, AI-driven business intelligence, executive dashboards, and contemporary data engineering techniques can be leveraged to build an analytics platform that can aid in making sponsorships decisions.

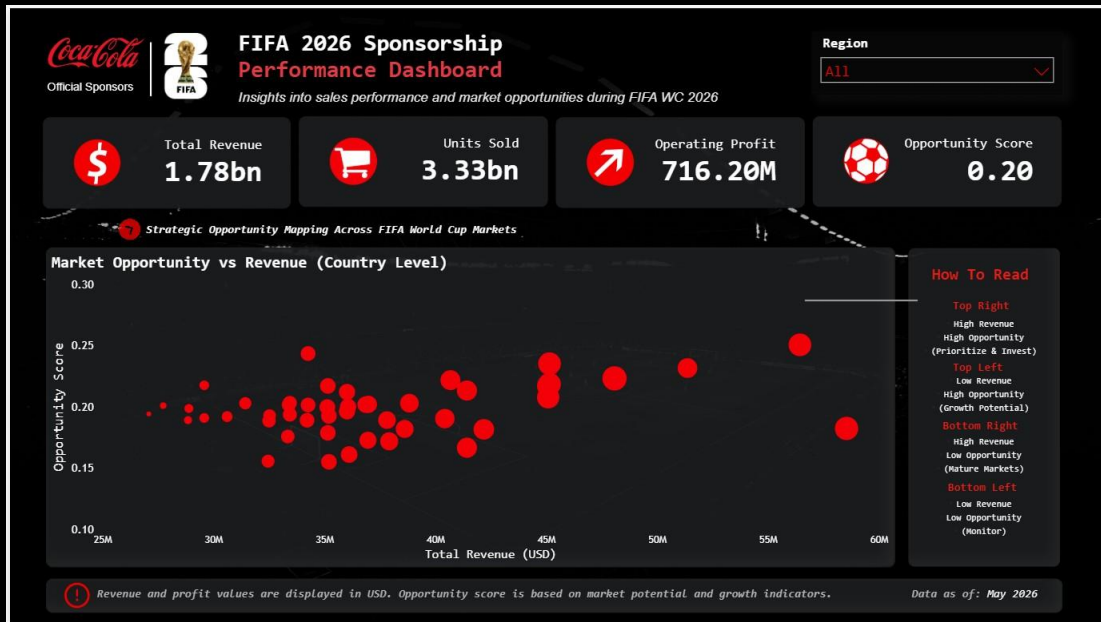


Fig. FIFA 2026 Sponsorship Performance Dashboard

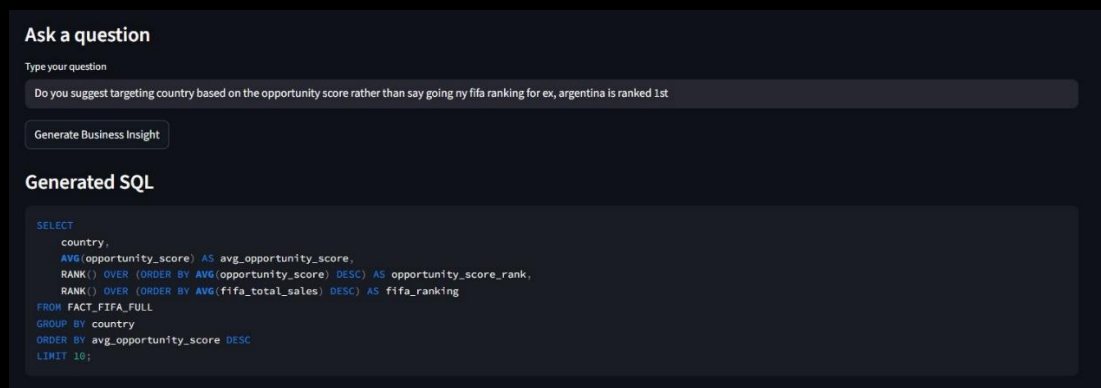


Fig. Sample Question on AI Assistant

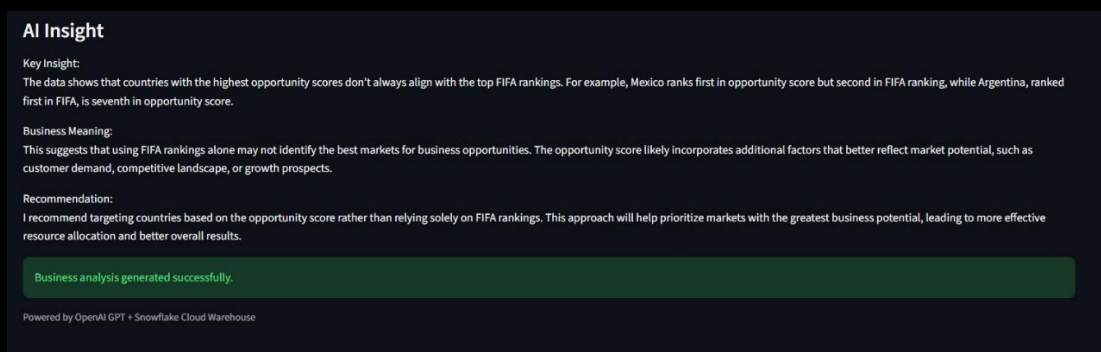


Fig. Sample Output on AI Assistant



Introduction

FIFA World Cup 2026 Sponsorship Analytics Platform is the end-to-end artificial intelligence-powered business intelligence platform designed to simulate real-world enterprise-level analytics workflow associated with sponsorship analysis and strategic market analysis.

The FIFA World Cup 2026 Sponsorship Analytics Project integrates cloud warehousing, ETL engineering, SQL analytics, executive dashboarding, and conversational AI into an analytical system capable of facilitating effective business intelligence.

This FIFA World Cup 2026 sponsorship analytics project was created for the purpose of analyzing sponsorship success, market opportunity in different regions, financial viability of operations, customer engagement, and strategic growth prospects in FIFA markets around the world.

To facilitate the creation of this scalable project that leverages modern approaches to analytics, this project utilizes Python-powered ETL pipelines, Snowflake cloud warehousing, Power BI executive dashboards, OpenAI-enabled SQL generation capabilities, and Streamlit conversational analytics.

In the course of this work, all aspects of the project are described, including data collection, ETL pipelines, Snowflake implementation, SQL analytics, executive dashboards, AI-integration capabilities, business insights gained, challenges faced, and possible enhancements to be introduced in the future.



Project Overview

The FIFA World Cup 2026 Sponsorship Analytics Platform is an end-to-end modern business intelligence and AI analytics ecosystem that is suitable for sponsorship strategy, market opportunities evaluation, and decision-making.

The project imitates the real enterprise analytics process by including several analytical modules into one coherent cloud-based system. Such a platform combines data engineering techniques, cloud-based data storage, SQL query engine, Power BI dashboards, and AI-assisted analytics into one solution for sponsors and their executives.

The analytical process starts with collecting multi-source datasets stored in different data files like CSV, XLSX, and JSON files. The collected data is preprocessed via several Python scripts creating an analytical dataset from these sources.

Finally, the preprocessed data is loaded into the cloud-based data warehouse – Snowflake database. This database works as a query engine in this project and serves as the common data source both for Power BI dashboards and Streamlit app.

The Power BI dashboard has been built to enable executive-level reports and analyses using KPIs, scatter plots, regional filtering, and tooltip interactions.

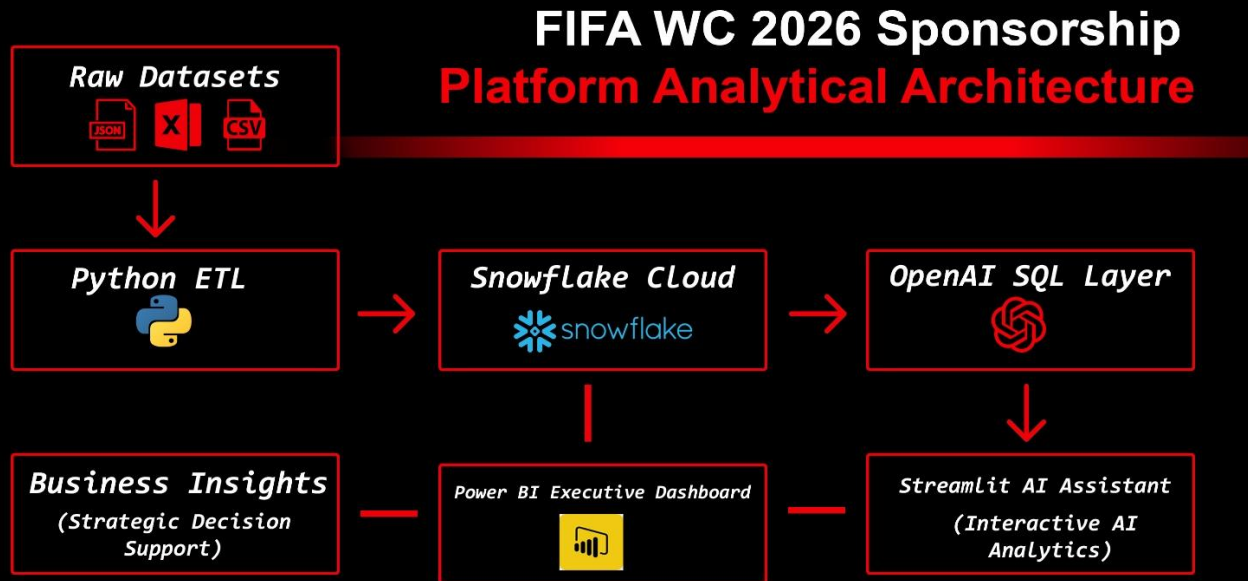


Fig. 2.1. FIFA WC 2026 Sponsorship Platform Analytical Architecture

To augment analysis outside of the visuals on the dashboard, an AI assistant was built using Streamlit and OpenAI GPT models. The assistant allows users to ask natural language questions about the business, automatically generates SQL queries, executes them using Snowflake, and provides explanations and recommendations powered by AI.

The entire system architecture shows how companies can leverage modern tools to build their analytics infrastructure.



Business Objectives

The key business goal for the FIFA World Cup 2026 Sponsorship Analytics Platform was to replicate an advanced sponsorship intelligence system for a modern business capable of making decisions based on analytical insights.

The FIFA World Cup and other global sports events attract a lot of sponsorships that are invested not in one but in many countries. Companies need to have effective analysis systems that can assess regional performance, customers' involvement, profitability, and market opportunities.

This platform aimed at answering such important business questions as:

- *In which countries are there the largest volume of sponsorship revenue?*
- *In which markets are there more growth potential?*
- *In which regions are there more opportunities even though less revenue is earned?*
- *How to find investment opportunities in FIFA markets?*
- *How AI analytics can facilitate accessibility for businesses and decision-making by executives?*

The platform incorporates financial KPIs, operational performance metrics, and opportunity scores to facilitate comparative market analysis in global FIFA sponsorship territories.

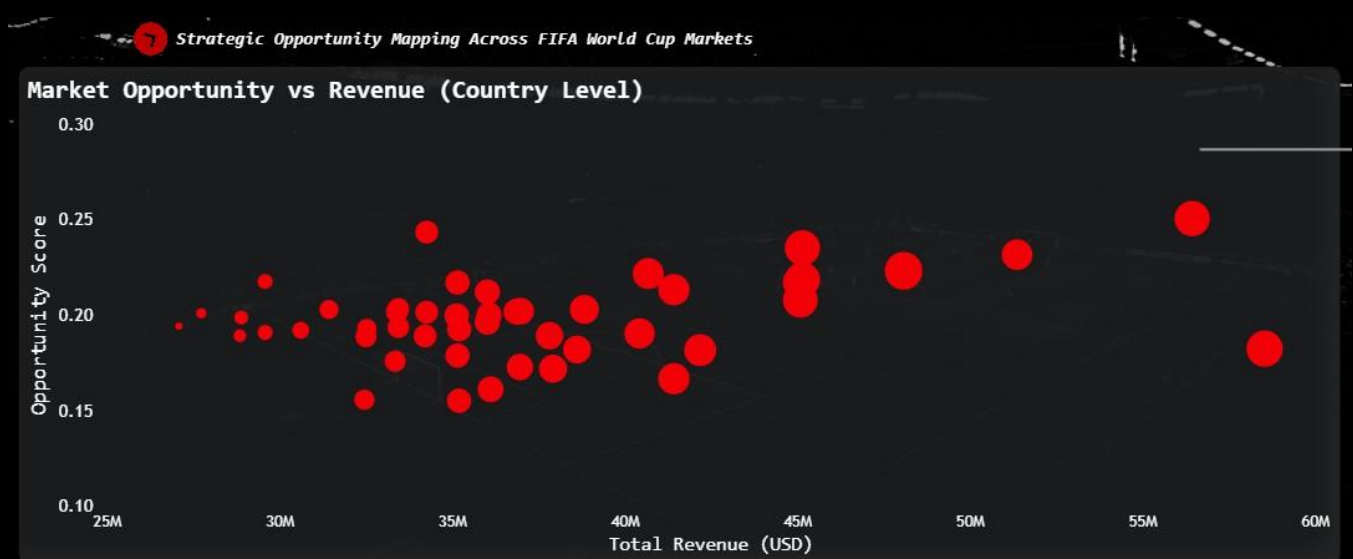


Fig. 3.1. Strategic Opportunity Mapping

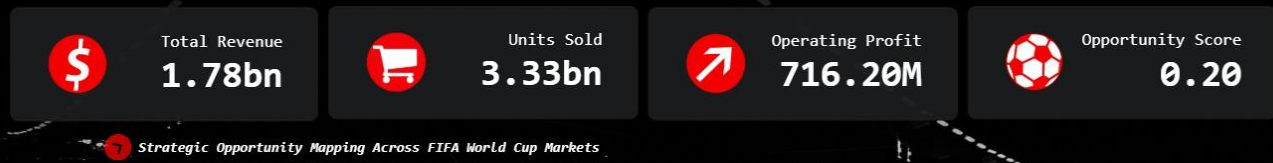


Fig. 3.2. Important KPI Cards for Business Insights

One of the primary goals of this business project has been to go further than static reporting and incorporate conversational artificial intelligence capabilities into the analytics process. This means that business users will be able to analyze sponsorship information by asking natural language questions, rather than manually crafting complex SQL queries or using dashboards alone.

Another aspect of the strategic focus of this project is the use of opportunity scoring metrics along with revenue analytics, which makes it possible for the organization to detect areas that might currently have relatively low revenue but represent very promising opportunities for the future.

In conclusion, the presented solution illustrates the way cloud analytics, dashboarding, and AI-assisted querying could all come together to boost sponsorships intelligence and strategic business decision-making.



Data Collection & ETL

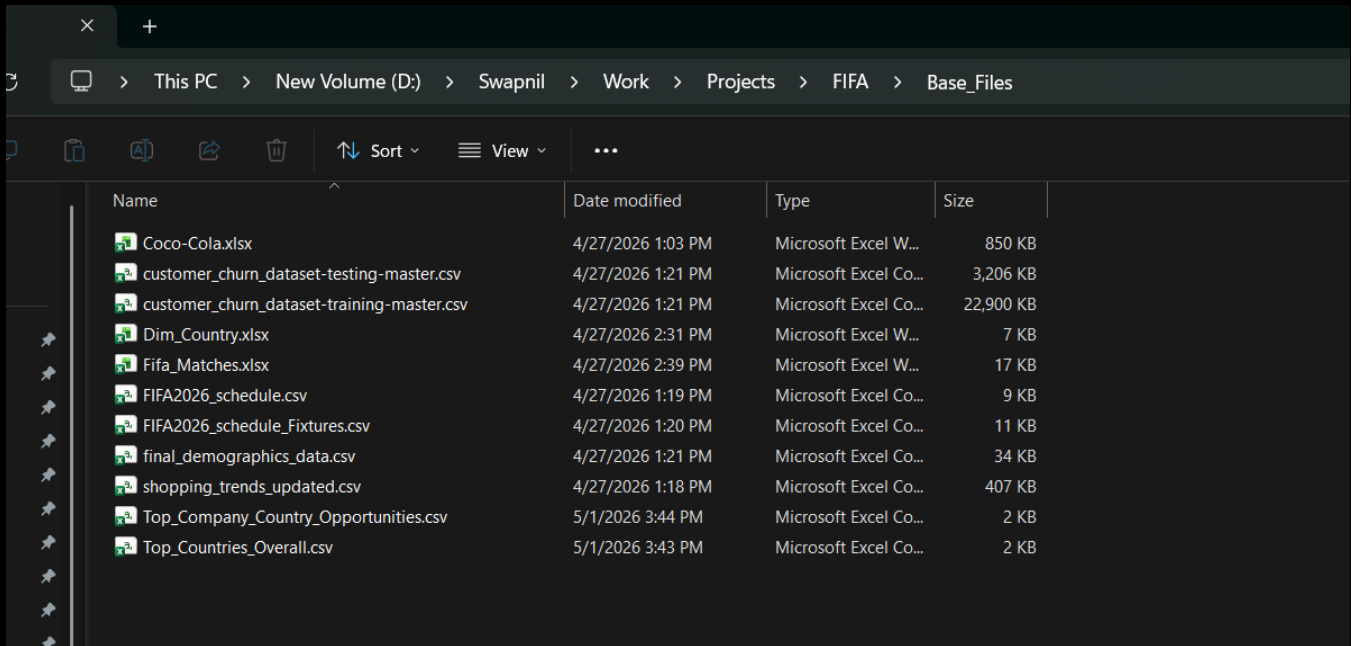
The FIFA World Cup 2026 Sponsorship Analytics Platform was built using multiple datasets collected from various structured and semi-structured data sources. The project utilized CSV, XLSX, and JSON files containing sponsorship performance metrics, regional sales data, customer engagement indicators, campaign analytics, and opportunity scoring information.

The use of multiple file formats was intended to simulate real-world enterprise analytical environments where business data is often distributed across heterogeneous systems and inconsistent structures.

To prepare the datasets for cloud analytics and business intelligence reporting, a Python-based ETL (Extract, Transform, Load) workflow was developed. Python libraries such as Pandas were used extensively for data ingestion, transformation, cleansing, normalization, and consolidation.

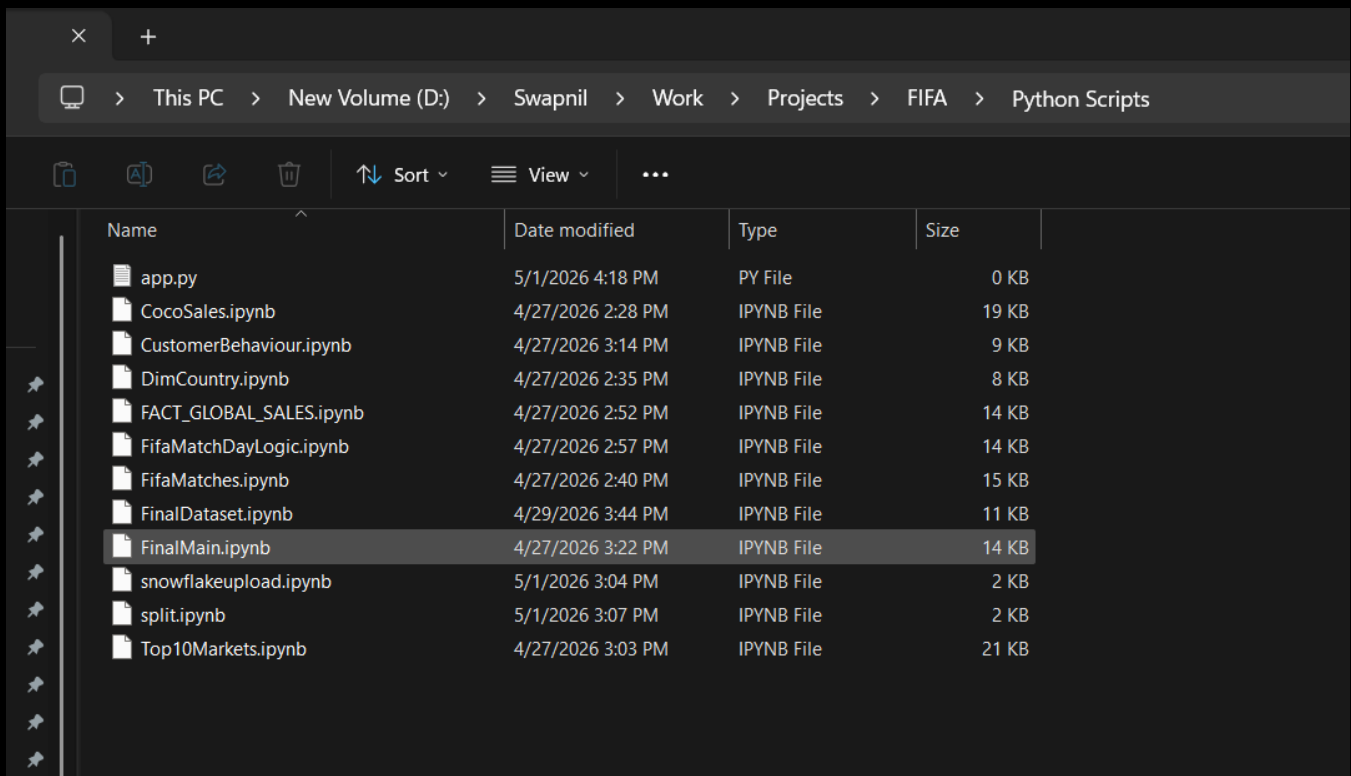
The ETL pipeline included:

- *File format conversion and standardization*
- *Header normalization*
- *Null value handling*
- *Duplicate record removal*
- *Datatype correction*
- *Country and region mapping*
- *KPI generation*
- *Revenue and profitability calculations*
- *Opportunity score integration*
- *Final analytical dataset consolidation*



Name	Date modified	Type	Size
Coco-Cola.xlsx	4/27/2026 1:03 PM	Microsoft Excel W...	850 KB
customer_churn_dataset-testing-master.csv	4/27/2026 1:21 PM	Microsoft Excel Co...	3,206 KB
customer_churn_dataset-training-master.csv	4/27/2026 1:21 PM	Microsoft Excel Co...	22,900 KB
Dim_Country.xlsx	4/27/2026 2:31 PM	Microsoft Excel W...	7 KB
Fifa_Matches.xlsx	4/27/2026 2:39 PM	Microsoft Excel W...	17 KB
FIFA2026_schedule.csv	4/27/2026 1:19 PM	Microsoft Excel Co...	9 KB
FIFA2026_schedule_Fixtures.csv	4/27/2026 1:20 PM	Microsoft Excel Co...	11 KB
final_demographics_data.csv	4/27/2026 1:21 PM	Microsoft Excel Co...	34 KB
shopping_trends_updated.csv	4/27/2026 1:18 PM	Microsoft Excel Co...	407 KB
Top_Company_Country_Opportunities.csv	5/1/2026 3:44 PM	Microsoft Excel Co...	2 KB
Top_Countries_Overall.csv	5/1/2026 3:43 PM	Microsoft Excel Co...	2 KB

Fig. 4.1. Base Files Directory



Name	Date modified	Type	Size
app.py	5/1/2026 4:18 PM	PY File	0 KB
CocoSales.ipynb	4/27/2026 2:28 PM	IPYNB File	19 KB
CustomerBehaviour.ipynb	4/27/2026 3:14 PM	IPYNB File	9 KB
DimCountry.ipynb	4/27/2026 2:35 PM	IPYNB File	8 KB
FACT_GLOBAL_SALES.ipynb	4/27/2026 2:52 PM	IPYNB File	14 KB
FifaMatchDayLogic.ipynb	4/27/2026 2:57 PM	IPYNB File	14 KB
FifaMatches.ipynb	4/27/2026 2:40 PM	IPYNB File	15 KB
FinalDataset.ipynb	4/29/2026 3:44 PM	IPYNB File	11 KB
FinalMain.ipynb	4/27/2026 3:22 PM	IPYNB File	14 KB
snowflakeupload.ipynb	5/1/2026 3:04 PM	IPYNB File	2 KB
split.ipynb	5/1/2026 3:07 PM	IPYNB File	2 KB
Top10Markets.ipynb	4/27/2026 3:03 PM	IPYNB File	21 KB

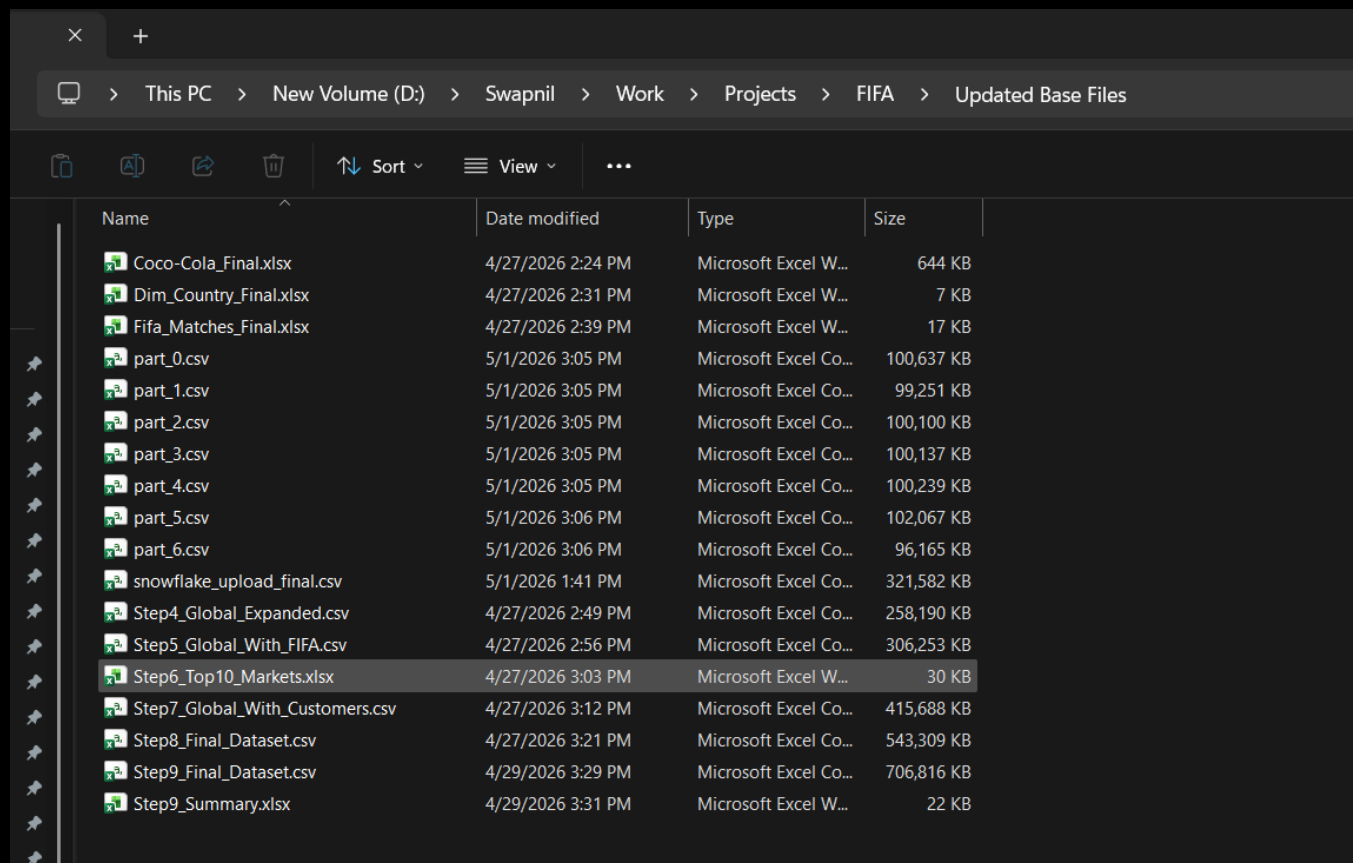
Fig. 4.2. Project Scripts Files Directory



The transformed datasets were merged into a centralized analytical dataset optimized for Snowflake cloud warehouse integration and downstream Power BI reporting.

The ETL workflow was intentionally designed to simulate scalable enterprise data engineering practices by separating raw source files from cleaned analytical outputs. This structure improved maintainability, reproducibility, and analytical consistency throughout the project lifecycle.

The final processed dataset served as the foundational analytical layer for Snowflake warehousing, SQL analytics, Power BI dashboarding, and AI-powered conversational querying.



Name	Date modified	Type	Size
Coco-Cola_Final.xlsx	4/27/2026 2:24 PM	Microsoft Excel W...	644 KB
Dim_Country_Final.xlsx	4/27/2026 2:31 PM	Microsoft Excel W...	7 KB
Fifa_Matches_Final.xlsx	4/27/2026 2:39 PM	Microsoft Excel W...	17 KB
part_0.csv	5/1/2026 3:05 PM	Microsoft Excel Co...	100,637 KB
part_1.csv	5/1/2026 3:05 PM	Microsoft Excel Co...	99,251 KB
part_2.csv	5/1/2026 3:05 PM	Microsoft Excel Co...	100,100 KB
part_3.csv	5/1/2026 3:05 PM	Microsoft Excel Co...	100,137 KB
part_4.csv	5/1/2026 3:05 PM	Microsoft Excel Co...	100,239 KB
part_5.csv	5/1/2026 3:06 PM	Microsoft Excel Co...	102,067 KB
part_6.csv	5/1/2026 3:06 PM	Microsoft Excel Co...	96,165 KB
snowflake_upload_final.csv	5/1/2026 1:41 PM	Microsoft Excel Co...	321,582 KB
Step4_Global_Expanded.csv	4/27/2026 2:49 PM	Microsoft Excel Co...	258,190 KB
Step5_Global_With_FIFA.csv	4/27/2026 2:56 PM	Microsoft Excel Co...	306,253 KB
Step6_Top10_Markets.xlsx	4/27/2026 3:03 PM	Microsoft Excel W...	30 KB
Step7_Global_With_Customers.csv	4/27/2026 3:12 PM	Microsoft Excel Co...	415,688 KB
Step8_Final_Dataset.csv	4/27/2026 3:21 PM	Microsoft Excel Co...	543,309 KB
Step9_Final_Dataset.csv	4/29/2026 3:29 PM	Microsoft Excel Co...	706,816 KB
Step9_Summary.xlsx	4/29/2026 3:31 PM	Microsoft Excel W...	22 KB

Fig. 4.3. Production Databases Files Directory (Final Datasets)



Snowflake Cloud Warehouse

Snowflake was chosen as the centralized cloud data warehouse used for the FIFA World Cup 2026 Sponsorship Analytics Platform. Snowflake worked as the central analytical database layer, responsible for storing, processing, and providing data to both Power BI dashboards and the intelligent Streamlit analytics application.

The use of Snowflake made it possible to create a modern enterprise cloud analytics solution, capable of executing scalable analytics, centralized reporting, and assisted AI-based queries.

The Snowflake database structure contained:

- *FIFA_DB database*
- *MARKETING schema*
- *FACT_FIFA_FULL analytical fact table*

The screenshot displays the Snowflake web interface. On the left, the 'Database Explorer' sidebar shows the hierarchy: FIFA_DB > INFORMATION_SCHEMA > Views (61) > MARKETING > Tables (2) > FACT_FIFA > FACT_FIFA_FULL. The main panel shows the query results for the table 'FACT_FIFA_FULL'. The results are displayed in a table format with columns: created_on, name, database_name, schema_name, kind, comment, cluster_by, and rows. The first row shows the table was created on 2026-05-01 14:55:41, named FACT_FIFA, in the FIFA_DB database and MARKETING schema, and is a TABLE.

	created_on	name	database_name	schema_name	kind	comment	cluster_by	rows
1	2026-05-01 14:55:41	FACT_FIFA	FIFA_DB	MARKETING	TABLE			

Fig. 5.1. Snowflake cloud warehouse environment showing the FIFA_DB database, MARKETING schema, and FACT_FIFA_FULL analytical table.



The finalized transformed analytical data from the Python ETL workflow was uploaded to Snowflake to be stored and processed centrally.

The reasons for choosing Snowflake include its ability to provide:

- *Cloud-native, scalable data storage architecture*
- *High-performance analytical querying abilities*
- *Smooth integration with Power BI*
- *Integration with Python and Streamlit environments*
- *Centralized analytics*

The FACT_FIFA_FULL table was the core analytical layer used in the production of reports on sponsorships and contained measures associated with:

- *Sponsorship revenue*
- *Units sold*
- *Operating margins*
- *Opportunity scores*
- *Customer engagement*
- *Campaign statistics*
- *Regional performance*
- *Market segmentation*

Table

Chart

</

Fig. 5.2. Preview of the FACT_FIFA_FULL analytical fact table stored within Snowflake.



SQL queries performed through Snowflake allowed for the derivation of consolidated business intelligence, strategic opportunity analysis, and executive level metrics.

The centralized warehousing structure also facilitated the operation of the Power BI report as well as the AI assistant on a common analytical dataset.

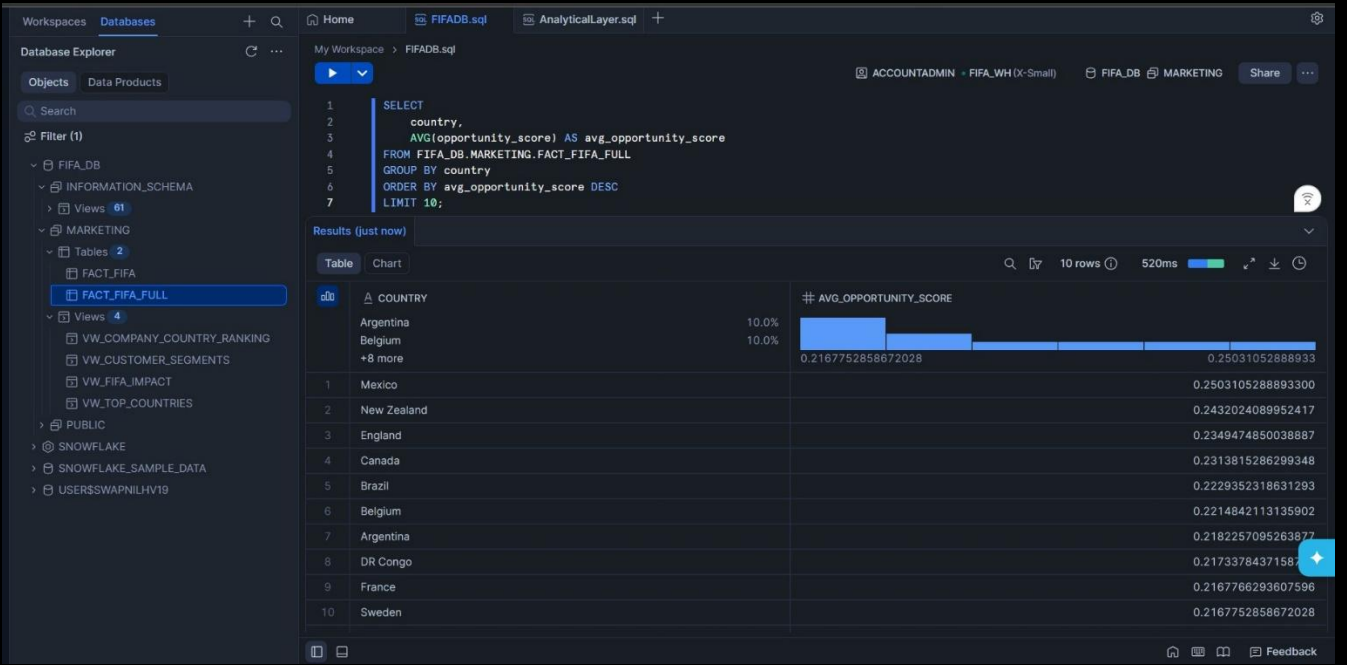


Fig. 5.3. SQL query execution within Snowflake used for sponsorship analytics and business reporting.

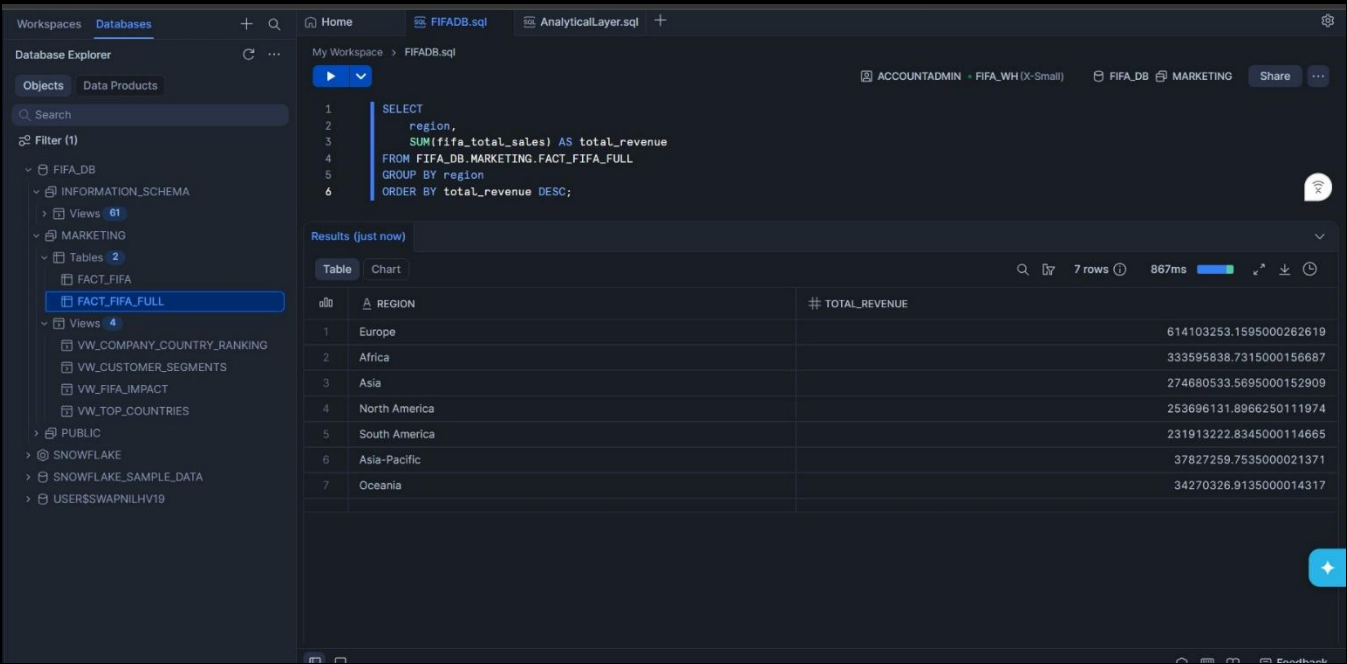


Fig. 5.4. Analytical SQL aggregation query used to identify top-performing FIFA sponsorship markets by revenue.



SQL Analytical Layer

SQL was heavily utilized within the Sponsorship Analytics Platform for the FIFA World Cup 2026 project to perform the analysis reporting, create KPIs, and build business intelligence workflows.

Specifically, SQL was utilized as the analytics layer right on top of the Snowflake cloud warehouse and acted as the main engine behind both Power BI and the AI-enabled Streamlit application.

Some analytical queries included:

- *Analysis of sponsorships financial performance*
- *Regional profitability calculation*
- *Evaluation of opportunity scores*
- *Segmentation of the market*
- *Calculation of customer engagement metrics*
- *Analysis of potential strategic sponsorships*

Additionally, within the context of an AI analytics platform, SQL was essential to allow dynamic execution of generated business queries by OpenAI on top of Snowflake warehouse.

Overall, the use case showcases one more way of implementing analytical logic via SQL within business intelligence solutions.

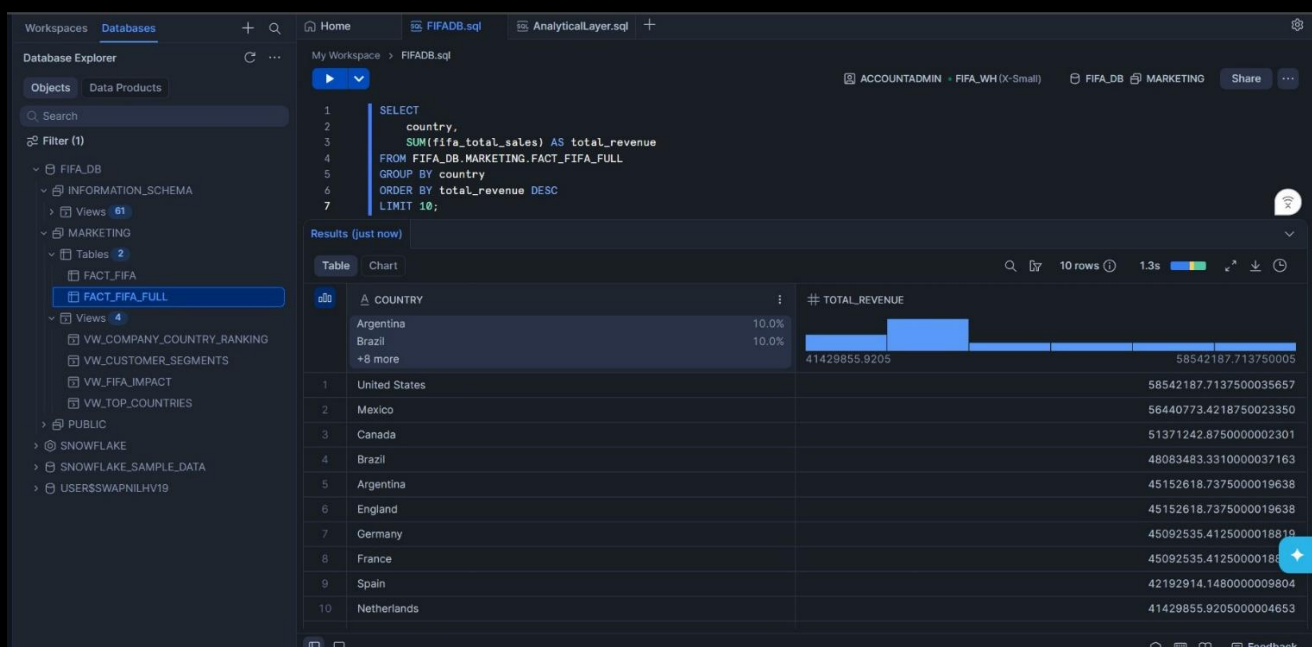


Fig. 6.1. SQL aggregation query executed within Snowflake for sponsorship revenue analysis across FIFA markets.



Power BI Dashboard

An extremely customized executive dashboard was created using Power BI to facilitate interactive sponsorship analytics, strategic market analysis, and executive business reporting for FIFA World Cup 2026 sponsorship intelligence.

The dashboard was created to resemble a modern business intelligence landscape of an enterprise, which primarily revolves around executive storytelling, KPI tracking, and market comparison.

The Power BI solution was implemented to connect seamlessly with the Snowflake data warehouse and leveraged the centralized FACT_FIFA_FULL analytical table as its core data source.

Some of the analytical capabilities that were created using the dashboard include:

- *Executive KPI summary cards*
- *Analysis of revenue and profit*
- *Visualization of opportunity scores*
- *Scatter plot analysis*
- *Regional filter analysis*
- *Analysis of top performing markets*
- *Dynamic tooltip interactions*
- *Sponsorship comparison reports*

The dashboard interface itself was deliberately designed using the dark enterprise theme in order to aid executive-level readability and analysis.

One of the key analytical visualizations that has been created as part of the dashboard is the scatter plot comparing sponsorship income with opportunity score within FIFA markets. As a result, it became possible to determine the most promising mature markets along with the emerging opportunities markets.

Interactive slicers and tooltip pages have been used in order to enhance analysis on regional segmentation of sponsorships.

The Power BI dashboard has been used as the main executive analytics interface within the system.

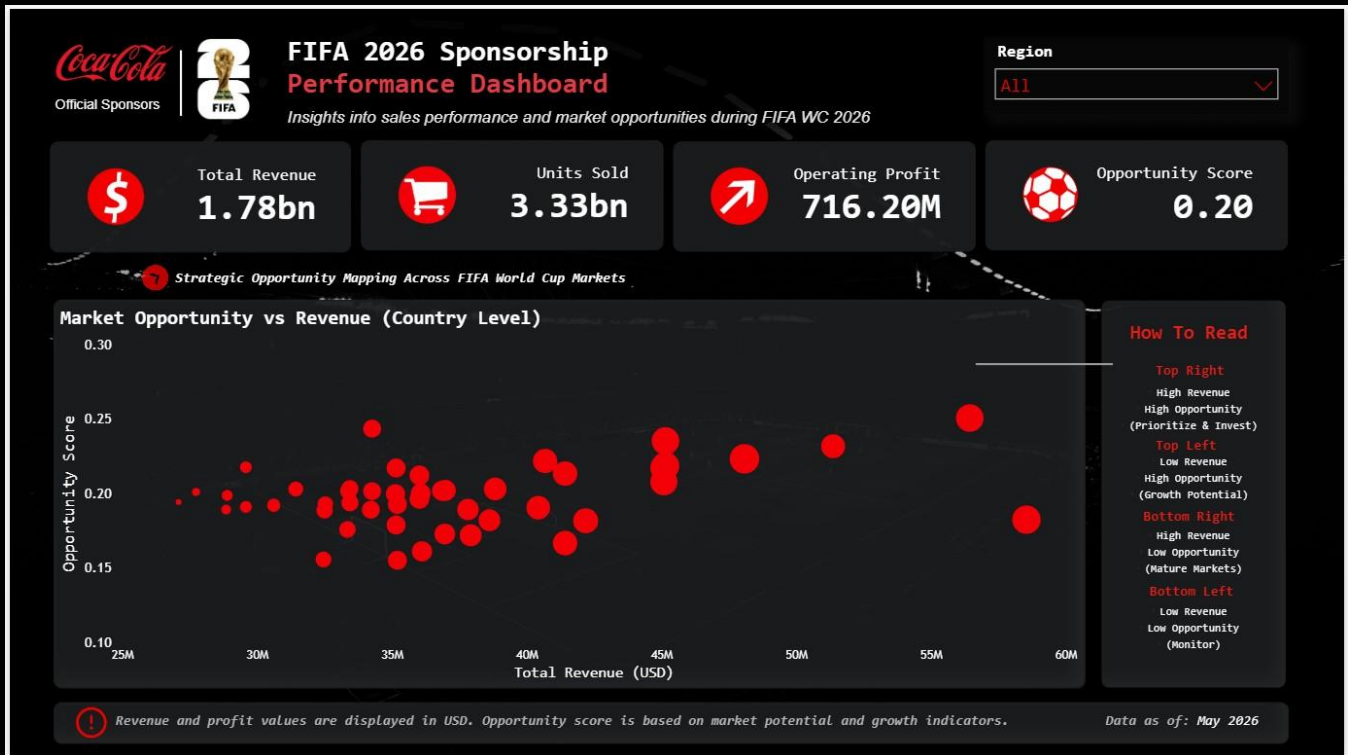


Fig. 7.1. Executive Power BI dashboard displaying FIFA World Cup 2026 sponsorship analytics and strategic market insights.

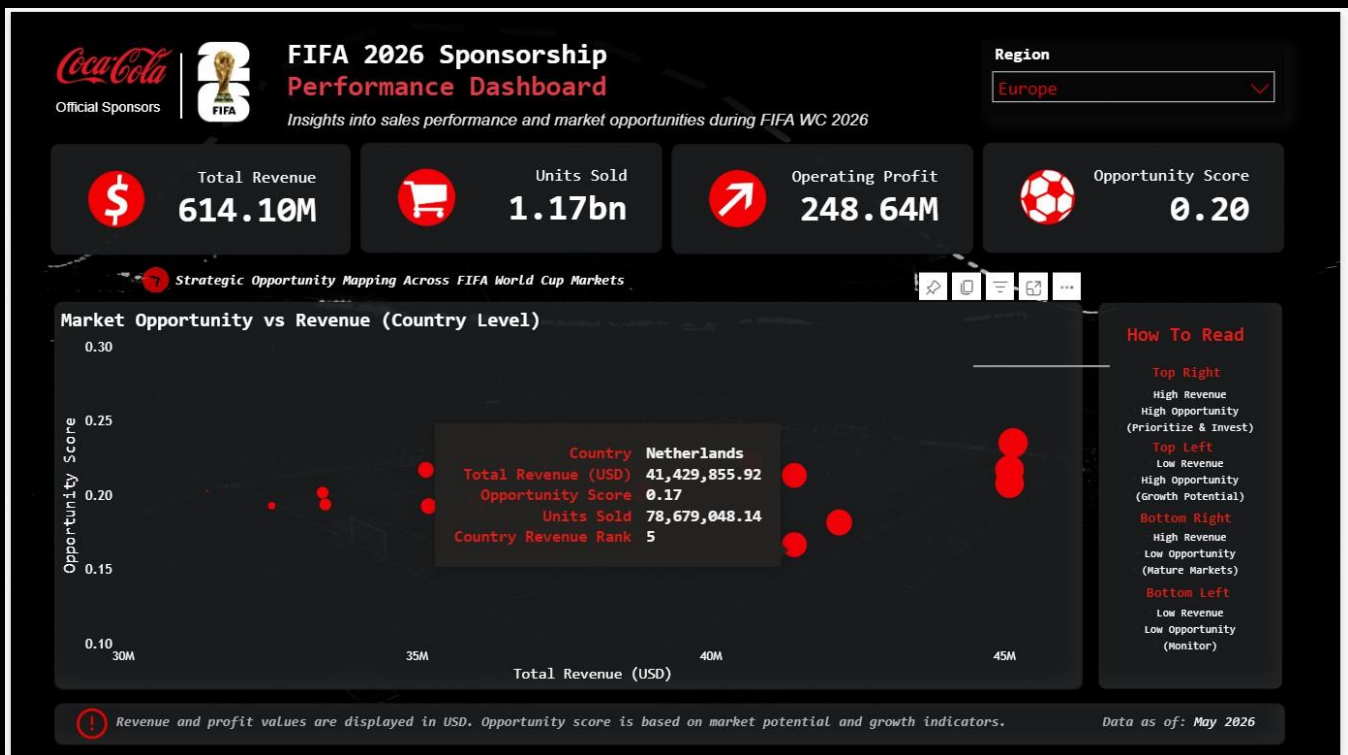


Fig. 7.2. Interactive tooltip visualization providing detailed sponsorship metrics for selected FIFA markets. (Filtered through Region)



Strategic Opportunity Mapping Across FIFA World Cup Markets

Fig. 7.3. Executive KPI summary cards displaying revenue, operating profit, units sold, and opportunity score metrics.

Strategic Opportunity Mapping Across FIFA World Cup Markets

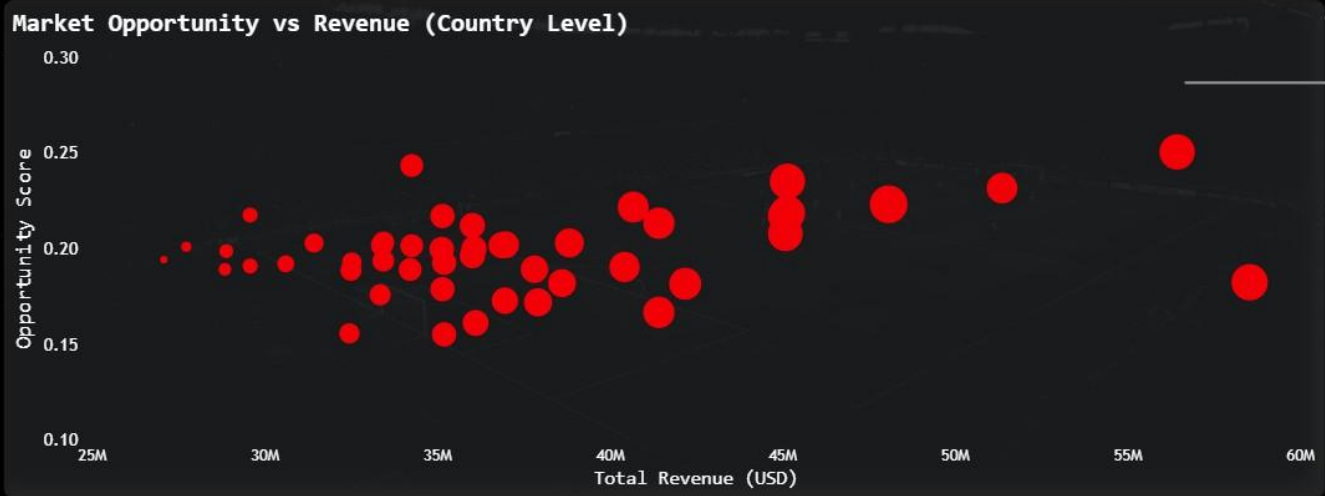


Fig. 7.4. Strategic market opportunity analysis comparing sponsorship revenue against opportunity score.



AI-Powered Analytics Assistant

To expand upon the limitations of conventional dashboards analysis, an AI-driven conversational analytics assistant was built with Streamlit, OpenAI GPT, and Snowflake connection.

The aim of the assistant was to allow the user to perform analysis of sponsorships through natural-language business questions without depending strictly on dashboards and/or SQL queries.

The Streamlit dashboard acted as the frontend tool that could connect with the Snowflake cloud data warehouse. OpenAI GPT models were used to generate SQL queries for the entered natural-language business questions by the user.

The work of the assistant included:

- *Natural language business question entry by the user*
- *Generation of SQL queries through AI*
- *Execution of dynamic SQL queries on Snowflake*
- *Getting analytical results back from the database*
- *Providing explanation and business recommendations through AI*

The AI-based assistant allowed asking for answers to questions regarding the following aspects:

- *Countries with highest sponsorships revenue*
- *Regions with the best opportunity score*
- *Markets with both highest revenue and high opportunity*
- *Sponsorship markets worth investment strategy*

Incorporation of OpenAI's technology for SQL generation made the system transition from being a conventional BI dash to a conversational analytics solution, which can be used for business exploration and decision-making purposes.

Some optimizations were added to ensure better reliability of SQL generation, schema knowledge, and consistency of generated analytics during the query processing.

The described implementation is a good example of how companies may use cloud analytics, artificial intelligence for queries, and conversational user interfaces in modern BI systems.

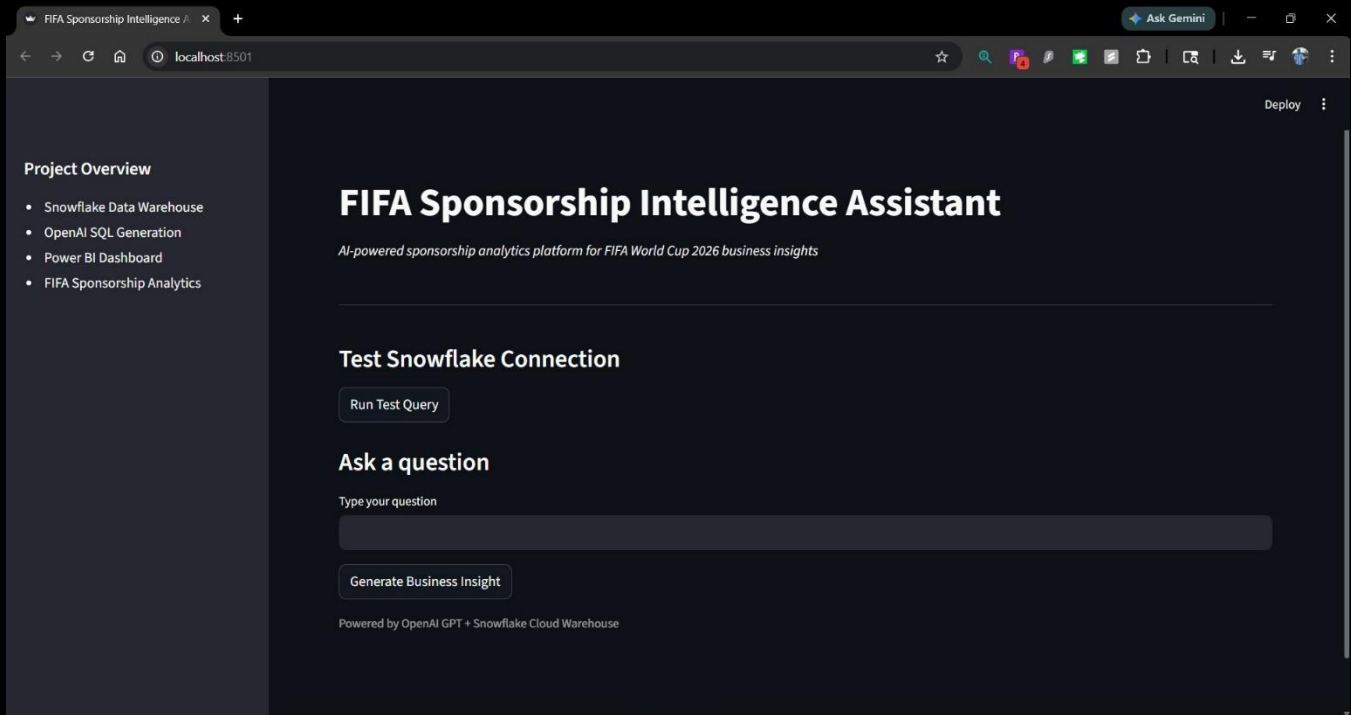


Fig. 8.1. Streamlit AI-powered sponsorship analytics assistant interface.

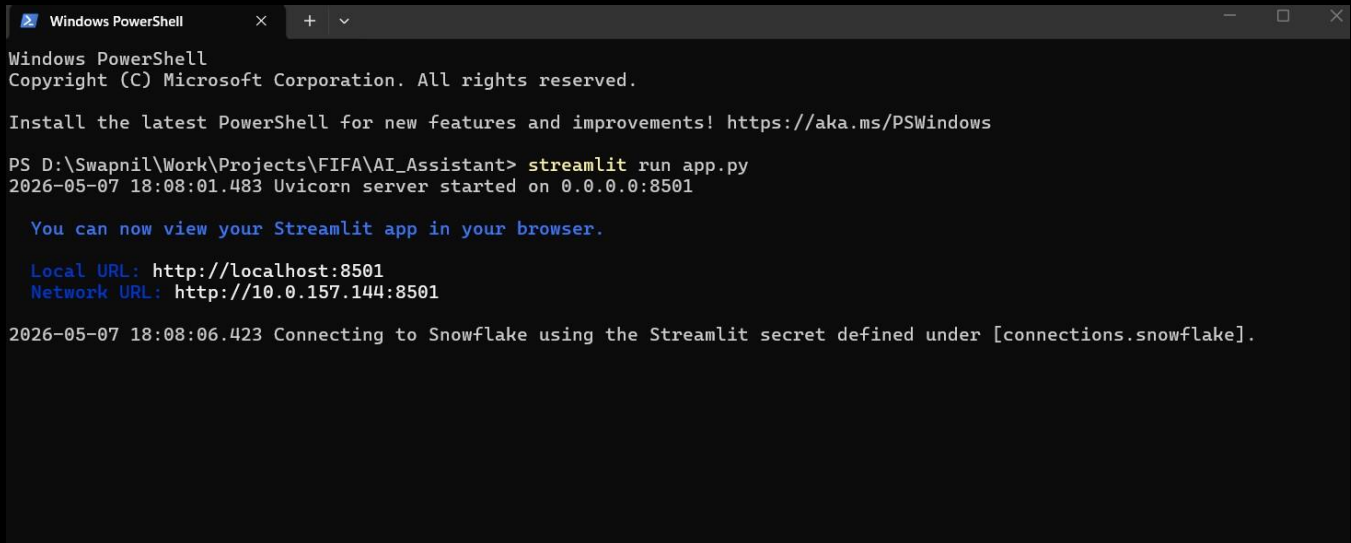


Fig. 8.2. Local deployment and execution of the Streamlit AI analytics application.



Ask a question

Type your question

which markets have the highest opportunity score

Generate Business Insight

Generated SQL

```
SELECT
  country,
  AVG(opportunity_score) AS avg_opportunity_score
FROM FACT_FIFA_FULL
GROUP BY country
ORDER BY avg_opportunity_score DESC
LIMIT 10;
```

Fig. 8.3. OpenAI-generated SQL query dynamically created from a natural-language business question.

Result

	COUNTRY	AVG OPPORTUNITY_SCORE
0	Mexico	0.2503
1	New Zealand	0.2432
2	England	0.2349
3	Canada	0.2314
4	Brazil	0.2229
5	Belgium	0.2215
6	Argentina	0.2182
7	DR Congo	0.2173
8	France	0.2168
9	Sweden	0.2168

Fig. 8.4. Analytical query results returned dynamically from Snowflake through the AI assistant.

AI Insight

Key Insight:

The markets with the highest opportunity scores are Mexico, New Zealand, and England, indicating these countries have the greatest potential for business growth.

Business Meaning:

This means that these countries show stronger demand, favorable conditions, or untapped potential compared to others, making them prime targets for expanding our products or services.

Recommendation:

Focus marketing and sales efforts on Mexico, New Zealand, and England to maximize growth opportunities. Consider allocating more resources and tailoring strategies specifically for these markets to capitalize on their high potential.

Business analysis generated successfully.

Powered by OpenAI GPT + Snowflake Cloud Warehouse

Fig. 8.5. AI-generated business insight and recommendation generated from sponsorship analytics data.



Business Insights

The FIFA World Cup 2026 Sponsorship Analytics Platform yielded many business insights through the adoption of cloud-based analytics, dashboards for executives, and AI-based business intelligence techniques.

The insights from the analysis proved that revenue from sponsorships was highly uneven between different FIFA regions and market segments, thus showing the need for region-specific approaches in sponsorship analysis.

Among the business insights that were gleaned throughout the analytical project:

- North America and Europe were found to be the best sponsors in terms of generating sponsorship revenue, mainly due to the maturity of their commercial markets and spending behaviors.
- Several emerging markets possessed high scores in terms of opportunities although their revenues were relatively low, which showed good potential for future growth.
- The opportunity score approach helped identify strategic markets more effectively than the approach based on sponsorship revenue.
- Analysis of scatter plot showing sponsorship income versus opportunity helped in identifying:
 - *Mature markets generating good sponsorship revenue*
 - *Emerging market areas with high opportunity score*
 - *Poor performing sponsorships*
 - *Areas for strategic investment*

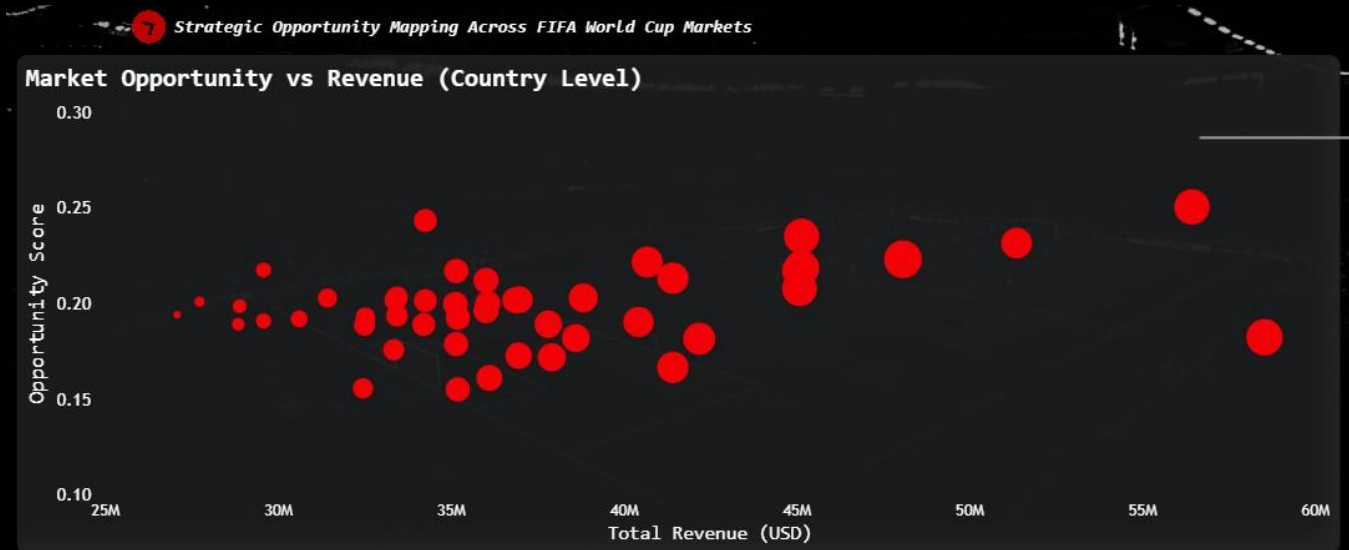


Fig. 9.1. Strategic market opportunity analysis comparing sponsorship revenue against opportunity score across FIFA markets.

- Analysis of KPIs showed the strong connection between customer engagement measures, campaign efficiency, and the generation of sponsorship revenue.
- Generation of SQL queries using artificial intelligence greatly facilitated analysis as the user could derive sponsorship analysis from business queries and not necessarily through SQL programming.



Challenges Faced

Several technical and analytical challenges arose during the development of the FIFA World Cup 2026 Sponsorship Analytics Platform. These challenges allowed for hands-on experience dealing with cloud integration and business intelligence using AI-assisted analytics tools.

Firstly, a challenge arose when standardizing and processing datasets from multiple file formats such as CSV, XLSX, and JSON files. Schema differences, different naming standards, and missing values had to be accounted for in the Python ETL process.

Secondly, a challenge that arose was developing an analytical database architecture capable of being used by both Power BI dashboards and AI-enabled SQL generation at the same time.

While developing an integration between Snowflake cloud warehouse and our analytics platform, there were numerous difficulties that appeared when validating table schema structures, query logic, and accessibility.

Creating the AI-driven analytics assistant was accompanied by several more challenges. Prompt engineering and schema-aware SQL generation posed difficulties because generated SQL sometimes contained invalid assumptions regarding the structure and datatypes used.

There were several user experience issues in the design of the dashboards that were also faced during the development process, which included:

- *Interactive tooltips*
- *Scatter plot labels*
- *Executive dashboard design*
- *Strategy visualization*
- *Slicers*
- *Dashboard responsiveness*

Maintaining a balance between complexity and readability for executives was a key aspect that emerged during the dashboard development process.

Another significant part of the development process involved integrating several technologies, such as Python, Snowflake, SQL, Power BI, OpenAI API, and Streamlit.

These experiences helped gain better insights into enterprise analytics processes, cloud computing deployment, AI-enabled query systems, and business intelligence engineering.



Future Improvements

Despite the success that the FIFA World Cup 2026 Sponsorship Analytics Platform has had in terms of showcasing cloud analytics, AI-supported business intelligence, and executive dashboards, there are multiple ways to improve the solution further in the future.

Possible future developments of the system may include:

- *Real-time APIs for sponsorship analytics and customer engagement tracking*
 - *Automated ETL orchestration and schedule-based cloud refresh processes*
 - *Predictive AI models for sponsorship revenue forecasting and trend analysis*
 - *AI recommendations for market target identification*
 - *Cloud installation of the Streamlit AI assistant*
 - *Role-based authorization and enhanced security controls*
 - *Executive alerts for real time*
 - *Conversational analytics with contextual AI memory*
- Advanced Power BI drill-through analytics and KPI prediction
 - Integration with scalable multi-cloud analytics infrastructure

At present, the solution has an excellent architecture foundation that can be used to expand the system's analytics capabilities and use AI in future.



Conclusion

The FIFA World Cup 2026 Sponsorship Analytics Platform successfully demonstrated the development of a modern end-to-end analytics ecosystem integrating ETL engineering, cloud warehousing, business intelligence reporting, and AI-assisted conversational analytics.

The project combined Python-based ETL workflows, Snowflake cloud infrastructure, SQL analytics, Power BI executive dashboards, OpenAI-powered SQL generation, and Streamlit application development into a unified analytical platform capable of supporting strategic sponsorship intelligence and executive decision-making.

The implementation demonstrated how cloud analytics, executive dashboarding, and AI-assisted querying can work together to improve sponsorship intelligence, strategic market evaluation, and executive decision-making workflows.

Through interactive dashboards, centralized cloud analytics, and AI-generated business insights, the platform showcased how organizations can improve analytical accessibility and business intelligence capabilities within modern enterprise environments.

The project also provided practical experience in data engineering, cloud analytics architecture, dashboard development, AI integration, prompt engineering, SQL analytics, and enterprise business intelligence implementation.

Overall, the final solution represents a scalable and portfolio-grade analytics platform capable of supporting real-world enterprise sponsorship intelligence workflows for FIFA World Cup 2026 business scenarios.





Appendix

Appendix A — Sample SQL Query

The following SQL query was used to identify the top-performing FIFA sponsorship markets based on total sponsorship revenue:

```
SELECT country,  
SUM(fifa_total_sales) AS total_sales  
FROM FACT_FIFA_FULL  
GROUP BY country  
ORDER BY total_sales DESC  
LIMIT 10;
```

Appendix B — AI Prompt Engineering Example

The AI-powered analytics assistant utilized structured prompt engineering techniques to improve SQL generation reliability and schema awareness during conversational analytics workflows.

Sample prompt structure:

Convert the following business question into a Snowflake SQL query using the FACT_FIFA_FULL table.

Rules:

- Return only SQL
- Use correct column names
- Apply aggregations when necessary
- Use GROUP BY for ranking queries
- Do not use markdown formatting

Appendix C — Key Analytical Columns



<i>Column Name</i>	<i>Description</i>
fifa_total_sales	Total sponsorship revenue
fifa_units_sold	Total units sold
fifa_operating_profit	Operating profitability
opportunity_score	Strategic market opportunity metric
ctr	Click-through rate
conversion_rate	Marketing conversion performance
roas	Return on advertising spend
high_opportunity	High-opportunity market indicator
region	FIFA regional market classification
country	Sponsorship market country

Appendix D — Core Technologies Used

<i>Technology</i>	<i>Purpose</i>
Python	ETL processing and data transformation
Snowflake	Cloud data warehousing
SQL	Analytical querying and KPI generation
Power BI	Executive dashboard development
Streamlit	AI analytics application development
OpenAI GPT	Natural-language SQL generation

Appendix E — Project Workflow Summary

